

Supplementary Analysis

Causal Band-Specialized Stacking for Multi-Horizon Solar Irradiance Forecasting

Journal revision — additional evidence

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source files : 47 existing result files

SCOPE

Every value in this document was computed from the manuscript's OWN saved outputs. No model was retrained, no architecture was changed, and no reported result was modified. These analyses add evidence to the submitted manuscript; they do not replace any part of it.

All values originate from executed computation.
Nothing is estimated, interpolated or fabricated.

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OK	Reported vs recomputed R^2 (consistency)
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S1_consistency_summary — Reported vs recomputed metrics from the m

source: S1_consistency_summary.csv

	metric	n	max_abs_diff	mean_abs_diff	tolerance	n_within_tolerance	all_agree
	fame_r2	42	4.899e-05	2.392e-05	0.0005	42	True
	fame_rmse_W	42	0.004875	0.002541	0.005	42	True
	fame_mae_W	42	0.004744	0.002349	0.005	42	True
	uxgb_r2	42	4.673e-05	2.449e-05	0.0005	42	True
	uxgb_rmse_W	42	0.004923	0.002357	0.005	42	True
	uxgb_mae_W	42	0.004794	0.002455	0.005	42	True

S2_consistency_perpair — Per-pair consistency verification (all 42 station

source: S2_consistency_perpair.csv

station	horizon	n_test	fame_r2_recomputed	fame_rmse_W_recomputed	fame_mae_W_recomputed	uxgb_r2_recomputed	uxgb_rmse_W_recomputed	uxgb_mae_W_recomputed
1	H1	5222	0.9503	76.22	39.26	0.9495	76.85	
1	H4	5221	0.9358	86.61	54.53	0.8972	109.7	
1	H8	5221	0.9141	100.2	71.83	0.8355	138.7	
1	H16	5220	0.8955	110.5	84.35	0.744	173	
1	H32	5217	0.912	101.5	74.71	0.5958	217.5	
1	H96	5208	0.8967	109.8	85.88	0.1746	310.5	
2	H1	5056	0.9558	46.24	24.63	0.9577	45.22	
2	H4	5056	0.9439	52.11	34.99	0.9084	66.57	
2	H8	5055	0.9398	53.97	37.23	0.8728	78.46	
2	H16	5054	0.9326	57.09	41.26	0.8191	93.55	
2	H32	5052	0.9305	57.96	42.01	0.7635	106.9	
2	H96	5042	0.929	58.6	42.08	0.4117	168.7	
4	H1	4849	0.9533	53.23	30.62	0.954	52.81	
4	H4	4848	0.9428	58.88	38.76	0.8773	86.25	
4	H8	4848	0.9389	60.87	41.23	0.789	113.1	
4	H16	4847	0.9337	63.41	44.65	0.5878	158.1	
4	H32	4844	0.931	64.72	46.49	0.2469	213.8	
4	H96	4835	0.9311	64.59	45.21	-0.1033	258.5	
5	H1	5004	0.91	89.73	50.06	0.9046	92.36	
5	H4	5004	0.8984	95.29	58.89	0.8191	127.2	
5	H8	5003	0.884	101.8	68.59	0.6989	164	
5	H16	5002	0.8752	105.6	72.6	0.417	228.2	
5	H32	5000	0.8627	110.7	75.67	0.3432	242	
5	H96	4990	0.862	110.7	76.4	-0.01327	300	
6	H1	5118	0.9229	101.2	56.45	0.9202	103	
6	H4	5118	0.9126	107.7	67.48	0.8135	157.4	
6	H8	5117	0.9042	112.8	74.62	0.7292	189.7	
6	H16	5116	0.8965	117.3	81.57	0.6153	226.1	
6	H32	5114	0.8951	118.1	81.3	0.5054	256.4	
6	H96	5104	0.8936	119	81.76	0.2461	316.8	
7	H1	4818	0.8206	111	67.91	0.8212	110.9	
7	H4	4818	0.8122	113.6	74.51	0.6666	151.4	
7	H8	4817	0.8065	115.3	78.19	0.6014	165.5	
7	H16	4816	0.7988	117.5	80.98	0.4977	185.7	

S2_consistency_perpair — Per-pair consistency verification (all 42 station

source: S2_consistency_perpair.csv

station	horizon	n_test	fame_r2_recomputed	fame_rmse_W_recomputed	fame_mae_W_recomputed	uxgb_r2_recomputed	uxgb_rmse_W_recomputed	uxgb_mae_W_recomputed
7	H32	4814	0.7976	117.8	81.62	0.3345	213.5	118.2
7	H96	4804	0.7902	119.3	81.23	0.1902	234.4	118.2
8	H1	4957	0.974	38.15	26.55	0.9766	36.16	118.2
8	H4	4957	0.9624	45.89	33.9	0.953	51.28	118.2
8	H8	4956	0.9538	50.84	39.14	0.901	74.45	118.2
8	H16	4955	0.9505	52.62	40.86	0.75	118.2	118.2
8	H32	4953	0.949	53.38	41.82	0.5318	161.8	118.2
8	H96	4943	0.947	54.47	42.53	0.1866	213.4	118.2

S3_consistency_aggregate — Per-horizon aggregate consistency against

source: S3_consistency_aggregate.csv

horizon	model	reported_mean_r2	recomputed_mean_r2	abs_diff	agrees
H1	FAME	0.9267	0.9267	1.3e-05	True
H1	Unified_XGB	0.9263	0.9263	4.1e-05	True
H4	FAME	0.9154	0.9155	6e-05	True
H4	Unified_XGB	0.8479	0.8479	3.6e-05	True
H8	FAME	0.9059	0.9059	1e-06	True
H8	Unified_XGB	0.7754	0.7754	9e-06	True
H16	FAME	0.8975	0.8975	3.9e-05	True
H16	Unified_XGB	0.633	0.633	1.4e-05	True
H32	FAME	0.8968	0.8968	3.5e-05	True
H32	Unified_XGB	0.4744	0.4744	3.8e-05	True
H96	FAME	0.8928	0.8928	1.2e-05	True
H96	Unified_XGB	0.1561	0.1561	1.7e-05	True

S4_implementation_audit — Implementation audit: executed correctness

source: S4_implementation_audit.csv

		check	status
A1		test-partition size consistent with a 70/15/15 split	PASS
A2		test size decreases monotonically with horizon	PASS
B1		causal decomposition is leakage-free	NOT APPLICABLE
C1		persistence equals the previous observed value	PASS
C2		persistence at the seam does not peek at the test block	PASS
D1		metric definitions match scikit-learn	PASS
E1		split-conformal attains nominal coverage with monotone width	PASS
E2		conformal split is chronological	PASS

longer horizons consume more rows to form the target, so the

calibra

S5_baseline_fairness — Baseline fairness audit (documented; baselines not rerun)

source: S5_baseline_fairness.csv

		check	status
F1 identical feature matrix and scaler for every model			VERIFIED
F2 identical chronological split for every model			VERIFIED
F3 identical optimiser and early-stopping protocol			VERIFIED
F4a sequence-length asymmetry		DISCLOSED	the LSTM band expert reads 48 steps of history; the three deep baselines
F4b epoch-budget asymmetry		DISCLOSED	
F4c evaluation-window asymmetry		DISCLOSED	
F6 baselines were not rerun for this revision		NOT APPLICABLE	

S6_paired_tests — Wilcoxon signed-rank, Cohen's d_z and 95% CIs, per h

source: S6_paired_tests.csv

metric	baseline	scope	n	mean_proposed	mean_baseline	mean_diff	ci95_lo	ci95_hi	wilcoxon_W	p_value	cohens_dz	effect_size	propos
skill	Unified	XGBoost	ALL	42	0.002869	-3.443	3.446	2.105	4.786	16	7.685e-11	0.8011	large
skill	Unified	XGBoost	H1	7	0.2682	0.2707	-0.002528	-0.03513	0.03007	12	0.8125	-0.07173	negligible
skill	Unified	XGBoost	H4	7	0.1177	-0.5635	0.6812	0.397	0.9654	0	0.01562	2.217	large
skill	Unified	XGBoost	H8	7	-0.002682	-1.453	1.451	0.9114	1.99	0	0.01562	2.488	large
skill	Unified	XGBoost	H16	7	-0.1037	-3.296	3.193	1.664	4.721	0	0.01562	1.932	large
skill	Unified	XGBoost	H32	7	-0.1073	-5.578	5.471	2.164	8.777	0	0.01562	1.53	large
skill	Unified	XGBoost	H96	7	-0.1549	-10.04	9.883	4.829	14.94	0	0.01562	1.809	large
r2	Unified	XGBoost	ALL	42	0.9059	0.6355	0.2704	0.1849	0.3558	12	3.183e-11	0.9862	large
r2	Unified	XGBoost	H1	7	0.9267	0.9263	0.000428	-0.002158	0.003014	12	0.8125	0.1531	negligible
r2	Unified	XGBoost	H4	7	0.9155	0.8479	0.0676	0.02541	0.1098	0	0.01562	1.482	large
r2	Unified	XGBoost	H8	7	0.9059	0.7754	0.1305	0.07244	0.1886	0	0.01562	2.079	large
r2	Unified	XGBoost	H16	7	0.8975	0.633	0.2646	0.1542	0.3749	0	0.01562	2.217	large
r2	Unified	XGBoost	H32	7	0.8968	0.4744	0.4224	0.2727	0.5721	0	0.01562	2.61	large
r2	Unified	XGBoost	H96	7	0.8928	0.1561	0.7367	0.5751	0.8984	0	0.01562	4.214	large
r2	Persistence	ALL	42	0.9059	0.9041	0.001782	-0.002984	0.006547	388	0.4348	0.1165	negligible	
r2	Persistence	H1	7	0.9267	0.9042	0.02251	0.01604	0.02898	0	0.01562	3.219	large	
r2	Persistence	H4	7	0.9155	0.9042	0.01128	0.004243	0.01832	0	0.01562	1.482	large	
r2	Persistence	H8	7	0.9059	0.9041	0.001763	-0.007162	0.01069	12	0.8125	0.1827	negligible	
r2	Persistence	H16	7	0.8975	0.9041	-0.006551	-0.01808	0.004982	6	0.2188	-0.5253	medium	
r2	Persistence	H32	7	0.8968	0.904	-0.007202	-0.0173	0.002891	6	0.2188	-0.66	medium	
r2	Persistence	H96	7	0.8928	0.9039	-0.01111	-0.02236	0.000147	4	0.1094	-0.9128	large	
rmse_W	Unified	XGBoost	ALL	42	84.68	156.9	-72.18	-91.29	-53.08	13	4.002e-11	-1.177	large
rmse_W	Unified	XGBoost	H1	7	73.69	73.89	-0.2011	-1.679	1.277	13	0.9375	-0.1259	negligible
rmse_W	Unified	XGBoost	H4	7	80.02	107.1	-27.09	-40.67	-13.51	0	0.01562	-1.844	large
rmse_W	Unified	XGBoost	H8	7	85.12	132	-46.87	-64.93	-28.8	0	0.01562	-2.4	large
rmse_W	Unified	XGBoost	H16	7	89.15	169	-79.84	-107.6	-52.06	0	0.01562	-2.659	large
rmse_W	Unified	XGBoost	H32	7	89.15	201.7	-112.6	-143.5	-81.64	0	0.01562	-3.368	large
rmse_W	Unified	XGBoost	H96	7	90.93	257.5	-166.5	-202.9	-130.2	0	0.01562	-4.232	large
rmse_W	Persistence	ALL	42	84.68	85.24	-0.5608	-2.902	1.781	416	0.6648	-0.07463	negligible	
rmse_W	Persistence	H1	7	73.69	85.26	-11.57	-14.96	-8.168	0	0.01562	-3.148	large	
rmse_W	Persistence	H4	7	80.02	85.26	-5.24	-7.619	-2.86	0	0.01562	-2.037	large	
rmse_W	Persistence	H8	7	85.12	85.26	-0.1434	-3.619	3.332	13	0.9375	-0.03816	negligible	
rmse_W	Persistence	H16	7	89.15	85.27	3.881	-1.831	9.592	4	0.1094	0.6284	medium	
rmse_W	Persistence	H32	7	89.15	85.25	3.899	-0.1079	7.906	4	0.1094	0.8999	large	

S6_paired_tests — Wilcoxon signed-rank, Cohen's d_z and 95% CIs, per h

source: S6_paired_tests.csv

metric	baseline	scope	n	mean_proposed	mean_baseline	mean_diff	ci95_lo	ci95_hi	wilcoxon_W	p_value	cohens_dz	effect_size	proposed_h
rmse_W	Persistence	H96	7	90.93	85.13	5.804	0.3946	11.21	1	0.03125	0.9923	large	
mae_W	Unified XGBoost	ALL	42	57.49	117.1	-59.61	-76.57	-42.65	14	5.002e-11	-1.095	large	
mae_W	Unified XGBoost	H1	7	42.21	42.34	-0.126	-1.578	1.326	14	1	-0.08022	negligible	
mae_W	Unified XGBoost	H4	7	51.87	70.53	-18.66	-30.58	-6.739	0	0.01562	-1.448	large	
mae_W	Unified XGBoost	H8	7	58.69	93.82	-35.13	-51.89	-18.37	0	0.01562	-1.939	large	
mae_W	Unified XGBoost	H16	7	63.75	126.9	-63.14	-88.63	-37.66	0	0.01562	-2.292	large	
mae_W	Unified XGBoost	H32	7	63.38	157.6	-94.26	-121.8	-66.68	0	0.01562	-3.161	large	
mae_W	Unified XGBoost	H96	7	65.01	211.3	-146.3	-177.2	-115.4	0	0.01562	-4.38	large	
mae_W	Persistence	ALL	42	57.49	54.6	2.889	-0.229	6.006	291	0.04464	0.2887	small	
mae_W	Persistence	H1	7	42.21	54.6	-12.39	-18.03	-6.746	0	0.01562	-2.03	large	
mae_W	Persistence	H4	7	51.87	54.6	-2.738	-6.326	0.8512	4	0.1094	-0.7055	medium	
mae_W	Persistence	H8	7	58.69	54.61	4.084	-0.242	8.409	0	0.01562	0.8731	large	
mae_W	Persistence	H16	7	63.75	54.61	9.145	2.563	15.73	0	0.01562	1.285	large	
mae_W	Persistence	H32	7	63.38	54.6	8.777	3.498	14.06	0	0.01562	1.538	large	
mae_W	Persistence	H96	7	65.01	54.56	10.45	3.13	17.78	0	0.01562	1.32	large	

S7_holm_bonferroni — Holm-Bonferroni correction across the per-horizon

source: S7_holm_bonferroni.csv

metric	baseline	scope	n	mean_proposed	mean_baseline	mean_diff	ci95_lo	ci95_hi	wilcoxon_W	p_value	cohens_dz	effect_size	proposed
mae_W	Persistence	H1	7	42.21	54.6	-12.39	-18.03	-6.746	0	0.01562	-2.03	large	
mae_W	Persistence	H4	7	51.87	54.6	-2.738	-6.326	0.8512	4	0.1094	-0.7055	medium	
mae_W	Persistence	H8	7	58.69	54.61	4.084	-0.242	8.409	0	0.01562	0.8731	large	
mae_W	Persistence	H16	7	63.75	54.61	9.145	2.563	15.73	0	0.01562	1.285	large	
mae_W	Persistence	H32	7	63.38	54.6	8.777	3.498	14.06	0	0.01562	1.538	large	
mae_W	Persistence	H96	7	65.01	54.56	10.45	3.13	17.78	0	0.01562	1.32	large	
mae_W	Unified XGBoost	H1	7	42.21	42.34	-0.126	-1.578	1.326	14	1	-0.08022	negligible	
mae_W	Unified XGBoost	H4	7	51.87	70.53	-18.66	-30.58	-6.739	0	0.01562	-1.448	large	
mae_W	Unified XGBoost	H8	7	58.69	93.82	-35.13	-51.89	-18.37	0	0.01562	-1.939	large	
mae_W	Unified XGBoost	H16	7	63.75	126.9	-63.14	-88.63	-37.66	0	0.01562	-2.292	large	
mae_W	Unified XGBoost	H32	7	63.38	157.6	-94.26	-121.8	-66.68	0	0.01562	-3.161	large	
mae_W	Unified XGBoost	H96	7	65.01	211.3	-146.3	-177.2	-115.4	0	0.01562	-4.38	large	
r2	Persistence	H1	7	0.9267	0.9042	0.02251	0.01604	0.02898	0	0.01562	3.219	large	
r2	Persistence	H4	7	0.9155	0.9042	0.01128	0.004243	0.01832	0	0.01562	1.482	large	
r2	Persistence	H8	7	0.9059	0.9041	0.001763	-0.007162	0.01069	12	0.8125	0.1827	negligible	
r2	Persistence	H16	7	0.8975	0.9041	-0.006551	-0.01808	0.004982	6	0.2188	-0.5253	medium	
r2	Persistence	H32	7	0.8968	0.904	-0.007202	-0.0173	0.002891	6	0.2188	-0.66	medium	
r2	Persistence	H96	7	0.8928	0.9039	-0.01111	-0.02236	0.000147	4	0.1094	-0.9128	large	
r2	Unified XGBoost	H1	7	0.9267	0.9263	0.000428	-0.002158	0.003014	12	0.8125	0.1531	negligible	
r2	Unified XGBoost	H4	7	0.9155	0.8479	0.0676	0.02541	0.1098	0	0.01562	1.482	large	
r2	Unified XGBoost	H8	7	0.9059	0.7754	0.1305	0.07244	0.1886	0	0.01562	2.079	large	
r2	Unified XGBoost	H16	7	0.8975	0.633	0.2646	0.1542	0.3749	0	0.01562	2.217	large	
r2	Unified XGBoost	H32	7	0.8968	0.4744	0.4224	0.2727	0.5721	0	0.01562	2.61	large	
r2	Unified XGBoost	H96	7	0.8928	0.1561	0.7367	0.5751	0.8984	0	0.01562	4.214	large	
rmse_W	Persistence	H1	7	73.69	85.26	-11.57	-14.96	-8.168	0	0.01562	-3.148	large	
rmse_W	Persistence	H4	7	80.02	85.26	-5.24	-7.619	-2.86	0	0.01562	-2.037	large	
rmse_W	Persistence	H8	7	85.12	85.26	-0.1434	-3.619	3.332	13	0.9375	-0.03816	negligible	
rmse_W	Persistence	H16	7	89.15	85.27	3.881	-1.831	9.592	4	0.1094	0.6284	medium	
rmse_W	Persistence	H32	7	89.15	85.25	3.899	-0.1079	7.906	4	0.1094	0.8999	large	
rmse_W	Persistence	H96	7	90.93	85.13	5.804	0.3946	11.21	1	0.03125	0.9923	large	
rmse_W	Unified XGBoost	H1	7	73.69	73.89	-0.2011	-1.679	1.277	13	0.9375	-0.1259	negligible	
rmse_W	Unified XGBoost	H4	7	80.02	107.1	-27.09	-40.67	-13.51	0	0.01562	-1.844	large	
rmse_W	Unified XGBoost	H8	7	85.12	132	-46.87	-64.93	-28.8	0	0.01562	-2.4	large	
rmse_W	Unified XGBoost	H16	7	89.15	169	-79.84	-107.6	-52.06	0	0.01562	-2.659	large	

S7_holm_bonferroni — Holm-Bonferroni correction across the per-horizon

source: S7_holm_bonferroni.csv

metric	baseline	scope	n	mean_proposed	mean_baseline	mean_diff	ci95_lo	ci95_hi	wilcoxon_W	p_value	cohens_dz	effect_size	proposed_b
rmse_W	Unified	XGBoost	H32	7	89.15	201.7	-112.6	-143.5	-81.64	0	0.01562	-3.368	large
rmse_W	Unified	XGBoost	H96	7	90.93	257.5	-166.5	-202.9	-130.2	0	0.01562	-4.232	large
skill	Unified	XGBoost	H1	7	0.2682	0.2707	-0.002528	-0.03513	0.03007	12	0.8125	-0.07173	negligible
skill	Unified	XGBoost	H4	7	0.1177	-0.5635	0.6812	0.397	0.9654	0	0.01562	2.217	large
skill	Unified	XGBoost	H8	7	-0.002682	-1.453	1.451	0.9114	1.99	0	0.01562	2.488	large
skill	Unified	XGBoost	H16	7	-0.1037	-3.296	3.193	1.664	4.721	0	0.01562	1.932	large
skill	Unified	XGBoost	H32	7	-0.1073	-5.578	5.471	2.164	8.777	0	0.01562	1.53	large
skill	Unified	XGBoost	H96	7	-0.1549	-10.04	9.883	4.829	14.94	0	0.01562	1.809	large

S8_bootstrap_ci — Bootstrap 95% confidence intervals on the skill score

source: S8_bootstrap_ci.csv

scope	model	n	mean_skill	boot_ci95_lo	boot_ci95_hi	n_boot
ALL	Proposed (FAME)	42	0.002869	-0.04907	0.05672	10000
ALL	Unified XGBoost	42	-3.443	-4.841	-2.195	10000
ALL	Persistence	42	0	0	0	10000
H1	Proposed (FAME)	7	0.2682	0.2058	0.3385	10000
H1	Unified XGBoost	7	0.2707	0.1942	0.3597	10000
H1	Persistence	7	0	0	0	10000
H4	Proposed (FAME)	7	0.1177	0.0809	0.1476	10000
H4	Unified XGBoost	7	-0.5635	-0.8049	-0.318	10000
H4	Persistence	7	0	0	0	10000
H8	Proposed (FAME)	7	-0.002682	-0.05776	0.04878	10000
H8	Unified XGBoost	7	-1.453	-1.896	-1.089	10000
H8	Persistence	7	0	0	0	10000
H16	Proposed (FAME)	7	-0.1037	-0.2086	-0.02105	10000
H16	Unified XGBoost	7	-3.296	-4.507	-2.21	10000
H16	Persistence	7	0	0	0	10000
H32	Proposed (FAME)	7	-0.1073	-0.1651	-0.03959	10000
H32	Unified XGBoost	7	-5.578	-8.288	-3.358	10000
H32	Persistence	7	0	0	0	10000
H96	Proposed (FAME)	7	-0.1549	-0.2429	-0.0688	10000
H96	Unified XGBoost	7	-10.04	-13.87	-6.441	10000
H96	Persistence	7	0	0	0	10000

S9_win_loss — Win/loss record against each baseline

source: S9_win_loss.csv

scope	baseline	n	wins	losses	ties	win_rate_pct
ALL	Unified XGBoost	42	38	4	0	90.48
ALL	Persistence	42	20	22	0	47.62
H1	Unified XGBoost	7	3	4	0	42.86
H1	Persistence	7	7	0	0	100
H4	Unified XGBoost	7	7	0	0	100
H4	Persistence	7	7	0	0	100
H8	Unified XGBoost	7	7	0	0	100
H8	Persistence	7	3	4	0	42.86
H16	Unified XGBoost	7	7	0	0	100
H16	Persistence	7	1	6	0	14.29
H32	Unified XGBoost	7	7	0	0	100
H32	Persistence	7	1	6	0	14.29
H96	Unified XGBoost	7	7	0	0	100
H96	Persistence	7	1	6	0	14.29

S10_conformal_summary — Split-conformal calibration and efficiency, all

source: S10_conformal_summary.csv

model	nominal_pct	n	mean_empirical_pct	min_empirical_pct	max_empirical_pct	mean_width_W	mean_width_per_sigma	mean_calibration_err
Persistence	90	42	92.99	85.71	97.15	279.4	0.9814	
Persistence	95	42	97.18	92.04	99.01	427.3	1.492	
Proposed (FAME)	90	42	93.38	83.27	97.11	287.8	1.006	
Proposed (FAME)	95	42	97.18	91.61	99.05	406.4	1.417	
Unified XGBoost	90	42	92.57	78.28	97.72	549.3	1.931	
Unified XGBoost	95	42	96.72	90.7	99.38	688.9	2.421	

S11_conformal_by_horizon — Conformal coverage and interval width by horizon

source: S11_conformal_by_horizon.csv

model	nominal_pct	horizon	mean_empirical_pct	mean_width_W	mean_calibration_error_pp	mean_winkler_W
Persistence	90	H1	93	279.4	3.002	375.4
Persistence	90	H16	93	279.5	2.999	375.6
Persistence	90	H32	93	279.5	2.997	375.6
Persistence	90	H4	93	279.4	3.002	375.4
Persistence	90	H8	93	279.6	3.002	375.6
Persistence	90	H96	92.94	278.9	2.94	375
Persistence	95	H1	97.18	427.6	2.179	507.6
Persistence	95	H16	97.18	427.7	2.178	507.7
Persistence	95	H32	97.18	427.3	2.177	507.6
Persistence	95	H4	97.18	427.6	2.179	507.6
Persistence	95	H8	97.18	427.7	2.179	507.6
Persistence	95	H96	97.18	426	2.177	506.2
Proposed (FAME)	90	H1	93.67	263.8	3.675	343.8
Proposed (FAME)	90	H16	92.82	293.8	2.825	376.1
Proposed (FAME)	90	H32	92.82	292.3	2.819	374.3
Proposed (FAME)	90	H4	93.63	279.8	3.633	356.9
Proposed (FAME)	90	H8	93.18	292.8	3.179	370.3
Proposed (FAME)	90	H96	94.18	304.4	4.179	375.5
Proposed (FAME)	95	H1	97.4	386.8	2.405	449.5
Proposed (FAME)	95	H16	96.96	413.4	1.96	485.2
Proposed (FAME)	95	H32	96.92	412.2	1.918	483.8
Proposed (FAME)	95	H4	97.3	397.1	2.3	461.5
Proposed (FAME)	95	H8	97.15	407	2.149	474.5
Proposed (FAME)	95	H96	97.33	421.7	2.33	487.7
Unified XGBoost	90	H1	93.33	262.9	3.326	346.1
Unified XGBoost	90	H16	91.49	596.2	1.491	735
Unified XGBoost	90	H32	92.78	705.7	2.782	820.2
Unified XGBoost	90	H4	92.61	388.3	2.614	499.2
Unified XGBoost	90	H8	91.52	467.2	1.524	598.4
Unified XGBoost	90	H96	93.66	875.4	3.66	978.6
Unified XGBoost	95	H1	97.15	381.7	2.15	451
Unified XGBoost	95	H16	95.84	727.4	0.8445	852.5
Unified XGBoost	95	H32	97.07	860.8	2.072	941.2
Unified XGBoost	95	H4	96.89	524.5	1.891	617.2

S11_conformal_by_horizon — Conformal coverage and interval width by h

source: S11_conformal_by_horizon.csv

	model	nominal_pct	horizon	mean_empirical_pct	mean_width_W	mean_calibration_error_pp	mean_winkler_W
Unified	XGBoost	95	H8	96.31	612.8	1.31	722.2
Unified	XGBoost	95	H96	97.07	1026	2.075	1104

S12_conformal_by_station — Conformal coverage and interval width by s

source: S12_conformal_by_station.csv

	model	nominal_pct	station	mean_empirical_pct	mean_width_W	mean_calibration_error_pp
	Persistence	90	1	86.16	203.7	-3.835
	Persistence	90	2	85.74	120.9	-4.263
	Persistence	90	4	95.75	196.7	5.749
	Persistence	90	5	96.32	339.7	6.321
	Persistence	90	6	97.15	432.6	7.146
	Persistence	90	7	96.72	484	6.718
	Persistence	90	8	93.1	178.1	3.097
	Persistence	95	1	95.69	378	0.6949
	Persistence	95	2	92.07	196.8	-2.93
	Persistence	95	4	99	316.1	3.996
	Persistence	95	5	98.53	562	3.527
	Persistence	95	6	98.79	636.6	3.788
	Persistence	95	7	98.88	677	3.878
	Persistence	95	8	97.3	224.6	2.295
	Proposed (FAME)	90	1	89.53	276.4	-0.4689
	Proposed (FAME)	90	2	85.64	129.5	-4.36
	Proposed (FAME)	90	4	95.88	208.3	5.88
	Proposed (FAME)	90	5	95.33	348.8	5.334
	Proposed (FAME)	90	6	96.91	420.8	6.911
	Proposed (FAME)	90	7	96.67	458.4	6.67
	Proposed (FAME)	90	8	93.73	172.6	3.729
	Proposed (FAME)	95	1	95.9	405.7	0.899
	Proposed (FAME)	95	2	92.49	198.8	-2.514
	Proposed (FAME)	95	4	98.65	298.3	3.652
	Proposed (FAME)	95	5	98.5	518.1	3.5
	Proposed (FAME)	95	6	98.67	599	3.671
	Proposed (FAME)	95	7	98.59	609.8	3.595
	Proposed (FAME)	95	8	97.44	215	2.437
	Unified XGBoost	90	1	90.82	546.9	0.8159
	Unified XGBoost	90	2	84.68	265.5	-5.321
	Unified XGBoost	90	4	94.68	524.7	4.676
	Unified XGBoost	90	5	92.41	659.4	2.412
	Unified XGBoost	90	6	95.66	784.7	5.659
	Unified XGBoost	90	7	96.68	682.3	6.684

S12_conformal_by_station — Conformal coverage and interval width by s

source: S12_conformal_by_station.csv

	model	nominal_pct	station	mean_empirical_pct	mean_width_W	mean_calibration_error_pp
Unified	XGBoost	90	8	93.04	381.6	3.037
Unified	XGBoost	95	1	96.68	757.1	1.679
Unified	XGBoost	95	2	93.03	364.9	-1.971
Unified	XGBoost	95	4	97.84	646.2	2.84
Unified	XGBoost	95	5	95.99	801.6	0.992
Unified	XGBoost	95	6	97.93	953.5	2.928
Unified	XGBoost	95	7	98.57	834.2	3.567
Unified	XGBoost	95	8	97.03	464.6	2.033

S13_conformal_crosscheck — Cross-check of the manuscript's Station-5 c

source: S13_conformal_crosscheck.csv

station	horizon	model	nominal_pct	reported_empirical	recomputed_empirical	reported_width	recomputed_width	coverage_absdiff	width_abs
5	H1	Proposed	90	96	96.04	342.6	342.6	0.0432	0.0
5	H1	Proposed	95	98.4	98.44	494.4	494.4	0.0412	0.0
5	H1	Unified XGB	90	96	96.04	341.6	341.6	0.0432	0.0
5	H1	Unified XGB	95	98.3	98.32	490.6	490.6	0.0213	0.0
5	H1	Persistence	90	96.3	96.32	340	340	0.0229	0.0
5	H1	Persistence	95	98.5	98.52	562	562	0.0212	0.0
5	H4	Proposed	90	95.4	95.36	345.6	345.6	0.0363	0.0
5	H4	Proposed	95	98.6	98.64	513.4	513.4	0.0411	0.0
5	H4	Unified XGB	90	95.1	95.12	477.1	477.1	0.0239	0.0
5	H4	Unified XGB	95	97.4	97.36	604.6	604.6	0.0379	0.0
5	H4	Persistence	90	96.3	96.32	340	340	0.0229	0.0
5	H4	Persistence	95	98.5	98.52	562	562	0.0212	0.0

S14_station_robustness — Robustness across the seven stations

source: S14_station_robustness.csv

station	n	mean_skill	std_skill	min_skill	max_skill	mean_r2	mean_rmse_W	mean_uxgb_skill	cv_abs	range	capacity_MW	beats_uxgb
1	6	-0.09045	0.288	-0.3782	0.3446	0.9174	97.48	-2.971	3.184	0.7228	50	True
2	6	0.0431	0.1582	-0.1055	0.3108	0.9386	54.33	-2.29	3.671	0.4163	130	True
4	6	-0.05517	0.1488	-0.186	0.1996	0.9385	60.95	-6.579	2.698	0.3856	130	True
5	6	-0.04369	0.1722	-0.2294	0.2036	0.882	102.3	-3.178	3.943	0.433	110	True
6	6	0.03704	0.1169	-0.07017	0.2253	0.9042	112.7	-2.635	3.157	0.2954	35	True
7	6	0.0894	0.0472	0.03504	0.1619	0.8043	115.8	-1.236	0.5279	0.1269	30	True
8	6	0.03985	0.2258	-0.1628	0.4315	0.9561	49.22	-5.212	5.665	0.5943	30	True

S15_horizon_robustness — Robustness across the six horizons

source: S15_horizon_robustness.csv

horizon	n	mean_skill	std_skill	min_skill	max_skill	mean_r2	mean_rmse_W	mean_uxgb_rmse_W	mean_pers_rmse_W	horizon_minutes	rmse_ratio_vs_uxgb
H1	7	0.2682	0.0969	0.1619	0.4315	0.9267	73.69	73.89	85.26	15	0.99
H4	7	0.1177	0.04933	0.02089	0.1775	0.9155	80.02	107.1	85.26	60	0.74
H8	7	-0.002682	0.07666	-0.1334	0.09708	0.9059	85.12	132	85.26	120	0.64
H16	7	-0.1037	0.1361	-0.3782	0.06163	0.8975	89.15	169	85.27	240	0.52
H32	7	-0.1073	0.0917	-0.2103	0.05804	0.8968	89.15	201.7	85.25	480	0.4
H96	7	-0.1549	0.1269	-0.3656	0.03504	0.8928	90.93	257.5	85.13	1440	0.35

S16_degradation — Skill degradation with lead time, per station

source: S16_degradation.csv

station	slope_per_horizon_step	intercept	r_value	p_value	stderr	skill_first	skill_last	total_degradation
1	-0.1357	0.2488	-0.8815	0.02025	0.03635	0.3446	-0.3656	0.7102
2	-0.08039	0.2441	-0.9505	0.003614	0.01314	0.3108	-0.1055	0.4163
4	-0.07513	0.1326	-0.9443	0.004566	0.01309	0.1996	-0.186	0.3856
5	-0.0908	0.1833	-0.9863	0.000282	0.007603	0.2036	-0.2294	0.433
6	-0.05955	0.1859	-0.9528	0.003291	0.009488	0.2253	-0.07017	0.2954
7	-0.02468	0.1511	-0.9783	0.000699	0.002611	0.1619	0.03504	0.1269
8	-0.1121	0.3201	-0.929	0.007385	0.02233	0.4315	-0.1628	0.5943

S17_failure_cases — Every station-horizon pair where the proposed model

source: S17_failure_cases.csv

station	horizon	fame_skill	fame_r2	uxgb_r2	fame_rmse_W	issues
1	H8	-0.1334	0.9141	0.8355	100.2	skill<0 (worse than persistence)
1	H16	-0.3782	0.8955	0.744	110.5	skill<0 (worse than persistence)
1	H32	-0.1639	0.912	0.5958	101.5	skill<0 (worse than persistence)
1	H96	-0.3656	0.8967	0.1746	109.8	skill<0 (worse than persistence)
2	H1	0.3108	0.9558	0.9577	46.24	R2 below unified XGBoost
2	H16	-0.05052	0.9326	0.8191	57.09	skill<0 (worse than persistence)
2	H32	-0.08208	0.9305	0.7635	57.96	skill<0 (worse than persistence)
2	H96	-0.1055	0.929	0.4117	58.6	skill<0 (worse than persistence)
4	H1	0.1996	0.9533	0.954	53.23	R2 below unified XGBoost
4	H8	-0.04655	0.9389	0.789	60.87	skill<0 (worse than persistence)
4	H16	-0.1358	0.9337	0.5878	63.41	skill<0 (worse than persistence)
4	H32	-0.1832	0.931	0.2469	64.72	skill<0 (worse than persistence)
4	H96	-0.186	0.9311	-0.1033	64.59	skill<0 (worse than persistence)
5	H8	-0.02557	0.884	0.6989	101.8	skill<0 (worse than persistence)
5	H16	-0.1023	0.8752	0.417	105.6	skill<0 (worse than persistence)
5	H32	-0.2103	0.8627	0.3432	110.7	skill<0 (worse than persistence)
5	H96	-0.2294	0.862	-0.01327	110.7	skill<0 (worse than persistence)
6	H16	-0.03926	0.8965	0.6153	117.3	skill<0 (worse than persistence)
6	H32	-0.05406	0.8951	0.5054	118.1	skill<0 (worse than persistence)
6	H96	-0.07017	0.8936	0.2461	119	skill<0 (worse than persistence)
7	H1	0.1619	0.8206	0.8212	111	R2 below unified XGBoost
8	H1	0.4315	0.974	0.9766	38.15	R2 below unified XGBoost
8	H8	-0.009434	0.9538	0.901	50.84	skill<0 (worse than persistence)
8	H16	-0.08181	0.9505	0.75	52.62	skill<0 (worse than persistence)
8	H32	-0.1158	0.949	0.5318	53.38	skill<0 (worse than persistence)
8	H96	-0.1628	0.947	0.1866	54.47	skill<0 (worse than persistence)

S18_error_structure — Residual bias, shape and autocorrelation by model

source: S18_error_structure.csv

station	horizon	model	bias_W	rmse_W	mae_W	mae_rmse_ratio	residual_std_W	residual_skew	residual_kurtosis	residual_autocorr_lag1
1	H1	Proposed (FAME)	3.484	76.22	39.26	0.5152	76.14	0.8744	16.33	0.0715
1	H1	Unified XGBoost	1.441	76.85	38.71	0.5037	76.83	0.7138	16.63	0.045
1	H1	Persistence	0.1369	94.15	63.66	0.6762	94.15	0.000412	11.21	0.0590
1	H4	Proposed (FAME)	7.704	86.61	54.53	0.6296	86.27	0.3774	8.918	0.315
1	H4	Unified XGBoost	2.735	109.7	65.74	0.5995	109.6	0.1822	7.711	0.60
1	H4	Persistence	0.1276	94.15	63.66	0.6762	94.15	0.000683	11.21	0.0590
1	H8	Proposed (FAME)	9.255	100.2	71.83	0.7166	99.81	0.2888	4.576	0.43
1	H8	Unified XGBoost	6.343	138.7	91.61	0.6606	138.5	0.2925	3.959	0.713
1	H8	Persistence	0.1276	94.15	63.66	0.6762	94.15	0.000683	11.21	0.0590
1	H16	Proposed (FAME)	11.69	110.5	84.35	0.7631	109.9	0.3291	3.021	0.492
1	H16	Unified XGBoost	-0.03786	173	125.4	0.7249	173	0.5963	1.973	0.825
1	H16	Persistence	0.1176	94.16	63.66	0.6761	94.16	0.000968	11.2	0.0587
1	H32	Proposed (FAME)	-0.5951	101.5	74.71	0.7362	101.5	0.4616	4.886	0.37
1	H32	Unified XGBoost	-35.48	217.5	165.3	0.7601	214.5	0.4498	1.427	0.890
1	H32	Persistence	0.08089	94.07	63.6	0.6761	94.07	-0.002714	11.25	0.0599
1	H96	Proposed (FAME)	5.49	109.8	85.88	0.782	109.7	0.152	3.475	0.427
1	H96	Unified XGBoost	-0.7199	310.5	257.1	0.8281	310.5	0.07701	-0.4909	0.930
1	H96	Persistence	0.004032	93.98	63.56	0.6763	93.98	-0.008239	11.29	0.0614
2	H1	Proposed (FAME)	4.622	46.24	24.63	0.5327	46.01	0.6806	14.23	0.00165
2	H1	Unified XGBoost	2.065	45.22	22.95	0.5074	45.17	0.9883	15.46	-0.00522
2	H1	Persistence	0.01725	55.7	37.1	0.6661	55.7	-0.2479	9.29	0.0487
2	H4	Proposed (FAME)	7.163	52.11	34.99	0.6714	51.61	0.1241	7.954	0.177
2	H4	Unified XGBoost	6.783	66.57	39.31	0.5906	66.22	0.8933	8.362	0.616
2	H4	Persistence	0.01725	55.7	37.1	0.6661	55.7	-0.2479	9.29	0.0487
2	H8	Proposed (FAME)	3.917	53.97	37.23	0.6899	53.83	0.04996	6.842	0.190
2	H8	Unified XGBoost	12.45	78.46	52.49	0.669	77.47	0.8236	4.15	0.715
2	H8	Persistence	0.00779	55.7	37.1	0.6661	55.7	-0.2476	9.291	0.048
2	H16	Proposed (FAME)	1.421	57.09	41.26	0.7226	57.08	-0.1694	5.803	0.239
2	H16	Unified XGBoost	17.96	93.55	67.38	0.7203	91.8	0.6458	2.495	0.787
2	H16	Persistence	0.002268	55.7	37.1	0.6661	55.7	-0.2473	9.289	0.0486
2	H32	Proposed (FAME)	-0.05788	57.96	42.01	0.7249	57.96	-0.2732	5.991	0.237
2	H32	Unified XGBoost	3.808	106.9	80.28	0.7508	106.9	0.3745	1.516	0.846
2	H32	Persistence	-0.000503	55.71	37.11	0.6661	55.71	-0.2471	9.285	0.0486
2	H96	Proposed (FAME)	2.815	58.6	42.08	0.718	58.53	-0.1754	5.783	0.222

S18_error_structure — Residual bias, shape and autocorrelation by model

source: S18_error_structure.csv

station	horizon	model	bias_W	rmse_W	mae_W	mae_rmse_ratio	residual_std_W	residual_skew	residual_kurtosis	residual_autocorr_lag1
2	H96	Unified XGBoost	4.726	168.7	138.3	0.8196	168.7	0.05184	-0.2777	0.9224
2	H96	Persistence	0.08092	55.74	37.11	0.6657	55.74	-0.2507	9.29	0.04756
4	H1	Proposed (FAME)	1.463	53.23	30.62	0.5752	53.21	0.5412	11.97	0.007879
4	H1	Unified XGBoost	2.739	52.81	30.07	0.5694	52.74	0.6181	11.75	-0.001837
4	H1	Persistence	0.01027	59.5	38.06	0.6396	59.5	0.644	10.54	0.04872
4	H4	Proposed (FAME)	2.066	58.88	38.76	0.6584	58.84	0.3843	8.325	0.2538
4	H4	Unified XGBoost	9.07	86.25	56.49	0.655	85.77	0.2233	4.962	0.6538
4	H4	Persistence	0.02081	59.5	38.06	0.6396	59.5	0.6437	10.54	0.04858
4	H8	Proposed (FAME)	3.486	60.87	41.23	0.6773	60.77	0.3081	7.36	0.2622
4	H8	Unified XGBoost	16.61	113.1	80.23	0.7093	111.9	0.05857	2.304	0.7729
4	H8	Persistence	0.02081	59.5	38.06	0.6396	59.5	0.6437	10.54	0.04858
4	H16	Proposed (FAME)	4.1	63.41	44.65	0.7041	63.28	0.2196	6.385	0.2715
4	H16	Unified XGBoost	23.18	158.1	120.1	0.7595	156.4	-0.2564	0.8177	0.8677
4	H16	Persistence	0.03163	59.5	38.05	0.6395	59.5	0.6434	10.54	0.04838
4	H32	Proposed (FAME)	1.522	64.72	46.49	0.7184	64.7	0.1526	5.562	0.2874
4	H32	Unified XGBoost	36.69	213.8	170.5	0.7977	210.6	-0.1609	0.02655	0.9269
4	H32	Persistence	0.07234	59.5	38.04	0.6393	59.5	0.6427	10.55	0.04766
4	H96	Proposed (FAME)	-7.426	64.59	45.21	0.7	64.16	0.1695	6.181	0.2587
4	H96	Unified XGBoost	28.42	258.5	216.4	0.8369	257	-0.3306	-0.5553	0.9453
4	H96	Persistence	0.1518	59.31	37.94	0.6397	59.31	0.6642	10.59	0.04533
5	H1	Proposed (FAME)	-1.374	89.73	50.06	0.5579	89.72	0.2823	10.89	0.1036
5	H1	Unified XGBoost	-1.997	92.36	53.05	0.5744	92.34	-0.09284	10.64	0.1187
5	H1	Persistence	0.1667	100.5	55.99	0.5568	100.5	0.163	13.09	-0.1941
5	H4	Proposed (FAME)	0.4354	95.29	58.89	0.618	95.29	-0.1561	7.657	0.1829
5	H4	Unified XGBoost	9.179	127.2	85.81	0.6747	126.9	-0.1554	4.231	0.6118
5	H4	Persistence	0.1667	100.5	55.99	0.5568	100.5	0.163	13.09	-0.1941
5	H8	Proposed (FAME)	9.907	101.8	68.59	0.6735	101.3	-0.3232	6.482	0.1983
5	H8	Unified XGBoost	27.39	164	121.2	0.739	161.7	-0.1518	1.529	0.7483
5	H8	Persistence	0.1731	100.6	55.99	0.5568	100.6	0.1628	13.09	-0.1941
5	H16	Proposed (FAME)	3.195	105.6	72.6	0.6876	105.5	-0.3671	6.009	0.1795
5	H16	Unified XGBoost	50.42	228.2	176.2	0.7723	222.5	-0.09393	0.3329	0.852
5	H16	Persistence	0.1783	100.6	56	0.5568	100.6	0.1626	13.08	-0.1941
5	H32	Proposed (FAME)	-10.99	110.7	75.67	0.6838	110.1	-0.3558	5.727	0.1995
5	H32	Unified XGBoost	14.09	242	195.3	0.8071	241.6	-0.3752	-0.1804	0.8903

S18_error_structure — Residual bias, shape and autocorrelation by model

source: S18_error_structure.csv

station	horizon	model	bias_W	rmse_W	mae_W	mae_rmse_ratio	residual_std_W	residual_skew	residual_kurtosis	residual_autocorr_lag1
5	H32	Persistence	0.1828	100.6	56.01	0.5569	100.6	0.1624	13.08	-0.1941
5	H96	Proposed (FAME)	-17.53	110.7	76.4	0.6901	109.3	-0.4689	5.593	0.1605
5	H96	Unified XGBoost	18.33	300	248.6	0.8287	299.4	-0.4716	-0.4882	0.9252
5	H96	Persistence	0.1727	99.85	55.8	0.5588	99.85	0.1838	12.96	-0.2002
6	H1	Proposed (FAME)	-1.623	101.2	56.45	0.5576	101.2	-0.1559	9.33	-0.01726
6	H1	Unified XGBoost	1.844	103	57.4	0.5573	103	-0.2649	9.618	-0.022
6	H1	Persistence	0.1589	115	71.13	0.6184	115	-0.04219	8.66	-0.007348
6	H4	Proposed (FAME)	-0.728	107.7	67.48	0.6263	107.7	-0.282	7.172	0.1599
6	H4	Unified XGBoost	4.362	157.4	104.7	0.665	157.4	-0.08375	3.469	0.6737
6	H4	Persistence	0.1589	115	71.13	0.6184	115	-0.04219	8.66	-0.007348
6	H8	Proposed (FAME)	5.527	112.8	74.62	0.6614	112.7	-0.2253	6.372	0.1831
6	H8	Unified XGBoost	7.906	189.7	135.9	0.716	189.6	-0.05241	1.759	0.777
6	H8	Persistence	0.1613	115	71.14	0.6184	115	-0.04225	8.658	-0.00734
6	H16	Proposed (FAME)	8.154	117.3	81.57	0.6955	117	-0.2673	5.746	0.1842
6	H16	Unified XGBoost	12.15	226.1	171	0.7561	225.8	-0.0315	0.7772	0.8195
6	H16	Persistence	0.1522	115	71.15	0.6184	115	-0.04202	8.656	-0.007482
6	H32	Proposed (FAME)	0.6807	118.1	81.3	0.6884	118.1	-0.2858	5.911	0.176
6	H32	Unified XGBoost	8.854	256.4	197.7	0.7708	256.3	0.03628	0.5599	0.8747
6	H32	Persistence	0.1122	115	71.13	0.6184	115	-0.0421	8.664	-0.007486
6	H96	Proposed (FAME)	14.9	119	81.76	0.6871	118.1	-0.3901	6.373	0.1475
6	H96	Unified XGBoost	36.01	316.8	254.5	0.8033	314.7	-0.205	-0.121	0.9114
6	H96	Persistence	0.08602	115	71.09	0.618	115	-0.04274	8.68	-0.007907
7	H1	Proposed (FAME)	10.65	111	67.91	0.6116	110.5	-0.1843	5.307	-0.1552
7	H1	Unified XGBoost	3.153	110.9	68.67	0.6194	110.8	-0.3201	4.816	-0.09689
7	H1	Persistence	0.1818	121.3	77.52	0.6391	121.3	0.02069	5.517	-0.1745
7	H4	Proposed (FAME)	6.991	113.6	74.51	0.6558	113.4	-0.09588	4.567	-0.005267
7	H4	Unified XGBoost	-6.624	151.4	103.7	0.6848	151.2	-0.1193	2.381	0.5875
7	H4	Persistence	0.1818	121.3	77.52	0.6391	121.3	0.02069	5.517	-0.1745
7	H8	Proposed (FAME)	13.57	115.3	78.19	0.6784	114.5	-0.228	4.569	-0.0175
7	H8	Unified XGBoost	-16.02	165.5	118.7	0.7173	164.7	-0.3649	1.677	0.669
7	H8	Persistence	0.1892	121.3	77.52	0.6391	121.3	0.02052	5.516	-0.1745
7	H16	Proposed (FAME)	17.96	117.5	80.98	0.689	116.1	-0.1897	4.367	-0.04257
7	H16	Unified XGBoost	13.6	185.7	137.4	0.74	185.2	-0.04213	0.8313	0.7385
7	H16	Persistence	0.1934	121.3	77.54	0.6391	121.3	0.02041	5.514	-0.1745

S18_error_structure — Residual bias, shape and autocorrelation by model

source: S18_error_structure.csv

station	horizon	model	bias_W	rmse_W	mae_W	mae_rmse_ratio	residual_std_W	residual_skew	residual_kurtosis	residual_autocorr_lag1
7	H32	Proposed (FAME)	3.432	117.8	81.62	0.693	117.7	-0.2343	4.129	-0.02243
7	H32	Unified XGBoost	25.81	213.5	169.5	0.7938	212	-0.008879	-0.04959	0.8006
7	H32	Persistence	0.2035	121.3	77.56	0.6392	121.3	0.02016	5.511	-0.1745
7	H96	Proposed (FAME)	-2.897	119.3	81.23	0.6807	119.3	-0.3284	4.469	-0.0498
7	H96	Unified XGBoost	23.5	234.4	190.6	0.8132	233.3	-0.238	-0.1444	0.8294
7	H96	Persistence	0.1924	121.5	77.7	0.6396	121.5	0.02041	5.494	-0.1745
8	H1	Proposed (FAME)	4.304	38.15	26.55	0.696	37.91	0.579	3.737	0.07923
8	H1	Unified XGBoost	3.473	36.16	25.53	0.7061	35.99	0.06049	1.964	-0.0255
8	H1	Persistence	0.1423	50.6	38.77	0.7662	50.6	0.02468	0.9006	0.0719
8	H4	Proposed (FAME)	6.557	45.89	33.9	0.7388	45.42	0.3771	2.496	0.3264
8	H4	Unified XGBoost	8.458	51.28	37.98	0.7405	50.58	0.1485	1.572	0.4751
8	H4	Persistence	0.1423	50.6	38.77	0.7662	50.6	0.02468	0.9006	0.0719
8	H8	Proposed (FAME)	4.277	50.84	39.14	0.77	50.66	-0.2718	0.7056	0.3171
8	H8	Unified XGBoost	14.12	74.45	56.69	0.7615	73.1	-0.05378	1.083	0.6762
8	H8	Persistence	0.1336	50.6	38.77	0.7662	50.6	0.02507	0.9009	0.07225
8	H16	Proposed (FAME)	3.957	52.62	40.86	0.7767	52.47	-0.3081	0.6951	0.2469
8	H16	Unified XGBoost	7.162	118.2	90.8	0.7679	118	-0.4579	0.768	0.8463
8	H16	Persistence	0.1526	50.59	38.76	0.7662	50.59	0.02527	0.9031	0.07359
8	H32	Proposed (FAME)	-0.8199	53.38	41.82	0.7833	53.38	-0.2691	0.4331	0.2449
8	H32	Unified XGBoost	-20.09	161.8	124.9	0.7717	160.6	-0.3374	0.3062	0.9165
8	H32	Persistence	0.1105	50.54	38.73	0.7664	50.54	0.02054	0.896	0.07313
8	H96	Proposed (FAME)	-0.9008	54.47	42.53	0.7807	54.47	-0.3377	0.6023	0.2685
8	H96	Unified XGBoost	9.187	213.4	173.9	0.8148	213.2	-0.1571	-0.09188	0.9335
8	H96	Persistence	0.03423	50.52	38.72	0.7665	50.52	0.01943	0.8981	0.07325

S19_conditional_skill — Performance stratified by irradiance regime

source: S19_conditional_skill.csv

regime	n_pairs	fame_rmse_W	uxgb_rmse_W	fame_skill
high	42	95.5	171	0.04271
low	42	57.93	147.1	-0.4703
medium	42	91.53	144.7	0.05181

S20_per_pair_metrics — All metrics for all models on all 42 station-horizon

source: S20_per_pair_metrics.csv

station	horizon	n_test	sigma_y_W	fame_r2	fame_rmse_W	fame_mae_W	fame_bias_W	fame_skill	unified_xgb_r2	unified_xgb_rmse_W	unified_xgb_mae_W
1	H1	5222	341.9	0.9503	76.22	39.26	3.484	0.3446	0.9495	76.85	38.73
1	H4	5221	341.9	0.9358	86.61	54.53	7.704	0.1538	0.8972	109.7	65.74
1	H8	5221	341.9	0.9141	100.2	71.83	9.255	-0.1334	0.8355	138.7	91.63
1	H16	5220	342	0.8955	110.5	84.35	11.69	-0.3782	0.744	173	125.4
1	H32	5217	342	0.912	101.5	74.71	-0.5951	-0.1639	0.5958	217.5	165.3
1	H96	5208	341.7	0.8967	109.8	85.88	5.49	-0.3656	0.1746	310.5	257.1
2	H1	5056	220	0.9558	46.24	24.63	4.622	0.3108	0.9577	45.22	22.95
2	H4	5056	220	0.9439	52.11	34.99	7.163	0.1247	0.9084	66.57	39.33
2	H8	5055	220	0.9398	53.97	37.23	3.917	0.06116	0.8728	78.46	52.49
2	H16	5054	219.9	0.9326	57.09	41.26	1.421	-0.05052	0.8191	93.55	67.38
2	H32	5052	219.9	0.9305	57.96	42.01	-0.05788	-0.08208	0.7635	106.9	80.28
2	H96	5042	220	0.929	58.6	42.08	2.815	-0.1055	0.4117	168.7	138.3
4	H1	4849	246.2	0.9533	53.23	30.62	1.463	0.1996	0.954	52.81	30.07
4	H4	4848	246.2	0.9428	58.88	38.76	2.066	0.02089	0.8773	86.25	56.49
4	H8	4848	246.2	0.9389	60.87	41.23	3.486	-0.04655	0.789	113.1	80.23
4	H16	4847	246.3	0.9337	63.41	44.65	4.1	-0.1358	0.5878	158.1	120.3
4	H32	4844	246.3	0.931	64.72	46.49	1.522	-0.1832	0.2469	213.8	170.5
4	H96	4835	246.2	0.9311	64.59	45.21	-7.426	-0.186	-0.1033	258.5	216.4
5	H1	5004	299	0.91	89.73	50.06	-1.374	0.2036	0.9046	92.36	53.05
5	H4	5004	299	0.8984	95.29	58.89	0.4354	0.1018	0.8191	127.2	85.83
5	H8	5003	298.9	0.884	101.8	68.59	9.907	-0.02557	0.6989	164	121.2
5	H16	5002	298.8	0.8752	105.6	72.6	3.195	-0.1023	0.417	228.2	176.3
5	H32	5000	298.6	0.8627	110.7	75.67	-10.99	-0.2103	0.3432	242	195.3
5	H96	4990	298	0.862	110.7	76.4	-17.53	-0.2294	-0.01327	300	248.6
6	H1	5118	364.6	0.9229	101.2	56.45	-1.623	0.2253	0.9202	103	57.4
6	H4	5118	364.6	0.9126	107.7	67.48	-0.728	0.1225	0.8135	157.4	104.7
6	H8	5117	364.6	0.9042	112.8	74.62	5.527	0.03793	0.7292	189.7	135.9
6	H16	5116	364.6	0.8965	117.3	81.57	8.154	-0.03926	0.6153	226.1	177.1
6	H32	5114	364.6	0.8951	118.1	81.3	0.6807	-0.05406	0.5054	256.4	197.7
6	H96	5104	364.8	0.8936	119	81.76	14.9	-0.07017	0.2461	316.8	254.5
7	H1	4818	262.2	0.8206	111	67.91	10.65	0.1619	0.8212	110.9	68.67
7	H4	4818	262.2	0.8122	113.6	74.51	6.991	0.1227	0.6666	151.4	103.7
7	H8	4817	262.1	0.8065	115.3	78.19	13.57	0.09708	0.6014	165.5	118.7
7	H16	4816	262	0.7988	117.5	80.98	17.96	0.06163	0.4977	185.7	137.4

S20_per_pair_metrics — All metrics for all models on all 42 station-horizon

source: S20_per_pair_metrics.csv

station	horizon	n_test	sigma_y_W	fame_r2	fame_rmse_W	fame_mae_W	fame_bias_W	fame_skill	unified_xgb_r2	unified_xgb_rmse_W	unified_xgb_mae_W
7	H32	4814	261.7	0.7976	117.8	81.62	3.432	0.05804	0.3345	213.5	169.5
7	H96	4804	260.5	0.7902	119.3	81.23	-2.897	0.03504	0.1902	234.4	190.6
8	H1	4957	236.6	0.974	38.15	26.55	4.304	0.4315	0.9766	36.16	25.53
8	H4	4957	236.6	0.9624	45.89	33.9	6.557	0.1775	0.953	51.28	37.98
8	H8	4956	236.6	0.9538	50.84	39.14	4.277	-0.009434	0.901	74.45	56.69
8	H16	4955	236.5	0.9505	52.62	40.86	3.957	-0.08181	0.75	118.2	90.8
8	H32	4953	236.5	0.949	53.38	41.82	-0.8199	-0.1158	0.5318	161.8	124.9
8	H96	4943	236.7	0.947	54.47	42.53	-0.9008	-0.1628	0.1866	213.4	173.9

S21_posthoc_fusion — Post-hoc fusion of saved predictions (oracle rows

source: S21_posthoc_fusion.csv

	variant	comparison_class	n	mean_r2	mean_rmse_W	mean_mae_W	mean_skill
[ORACLE]	convex(uxgb,pers), weight fitted on test-cal half	oracle	42	0.9299	67.33	45.61	0.07452
	[ORACLE] best_single per pair	oracle	42	0.9274	68.33	45.04	0.04968
	proposed (manuscript, no test access)	fair	42	0.9194	72.38	51.43	-0.07901
	equal_average(uxgb,pers)	fair	42	0.8648	91.39	67.85	-0.9972

S22_fusion_headtohead — Fair head-to-head among variants with no test

source: S22_fusion_headtohead.csv

	comparison	n	mean_rmse_diff_W	proposed_better_in	proposed_worse_in
proposed	vs equal_average(uxgb,pers)	42	-19.01	40	2

S23_proposed_vs_baselines — Manuscript model vs each baseline on the

source: S23_proposed_vs_baselines.csv

station	horizon	proposed_rmse_W	uxgb_rmse_W	pers_rmse_W	proposed_beats_uxgb	proposed_beats_pers
1	H1	69.88	70.81	88.68	True	True
1	H4	82.48	107.3	88.68	True	True
1	H8	99.23	136.7	88.68	True	False
1	H16	114.2	171	88.69	True	False
1	H32	100.9	218.5	88.69	True	False
1	H96	105	302	88.71	True	False
2	H1	49.26	48.3	58.05	False	True
2	H4	54.82	70.75	58.05	True	True
2	H8	57.68	87.75	58.05	True	True
2	H16	60.79	101.9	58.06	True	False
2	H32	61.11	107.3	58.07	True	False
2	H96	57.42	152.4	57.82	True	True
4	H1	38.29	38.24	45.29	False	True
4	H4	46.64	70.08	45.29	True	False
4	H8	50.03	102.8	45.29	True	False
4	H16	52.91	152	45.29	True	False
4	H32	53.65	203.5	45.3	True	False
4	H96	51.38	237.8	45.28	True	False
5	H1	67.24	71.79	74.96	True	True
5	H4	76.19	109.9	74.96	True	False
5	H8	84.54	152.8	74.96	True	False
5	H16	89.75	224.1	74.97	True	False
5	H32	97.45	238.5	74.97	True	False
5	H96	94.33	306.5	74.52	True	False
6	H1	68.93	70.42	87.6	True	True
6	H4	81.63	122.9	87.6	True	True
6	H8	88.78	158.4	87.6	True	False
6	H16	96.48	201	87.61	True	False
6	H32	96.96	213.2	87.63	True	False
6	H96	94.42	266.6	87.71	True	False
7	H1	80.58	79.95	89.51	False	True
7	H4	83.32	109.6	89.51	True	True
7	H8	87.59	122.4	89.51	True	True
7	H16	90.78	147.2	89.51	True	False

S23_proposed_vs_baselines — Manuscript model vs each baseline on the

source: S23_proposed_vs_baselines.csv

station	horizon	proposed_rmse_W	uxgb_rmse_W	pers_rmse_W	proposed_beats_uxgb	proposed_beats_pers
7	H32	91.43	190.5	89.46	True	False
7	H96	89.68	194.8	89.55	True	False
8	H1	33.33	33.73	47.27	True	True
8	H4	40.31	48.83	47.27	True	True
8	H8	46.93	71.91	47.28	True	True
8	H16	50.63	114.3	47.28	True	False
8	H32	53.09	147.6	47.28	True	False
8	H96	49.94	198.4	47.31	True	False

S24_scope_declaration — Which analyses are computable from frozen ou

source: S24_scope_declaration.csv

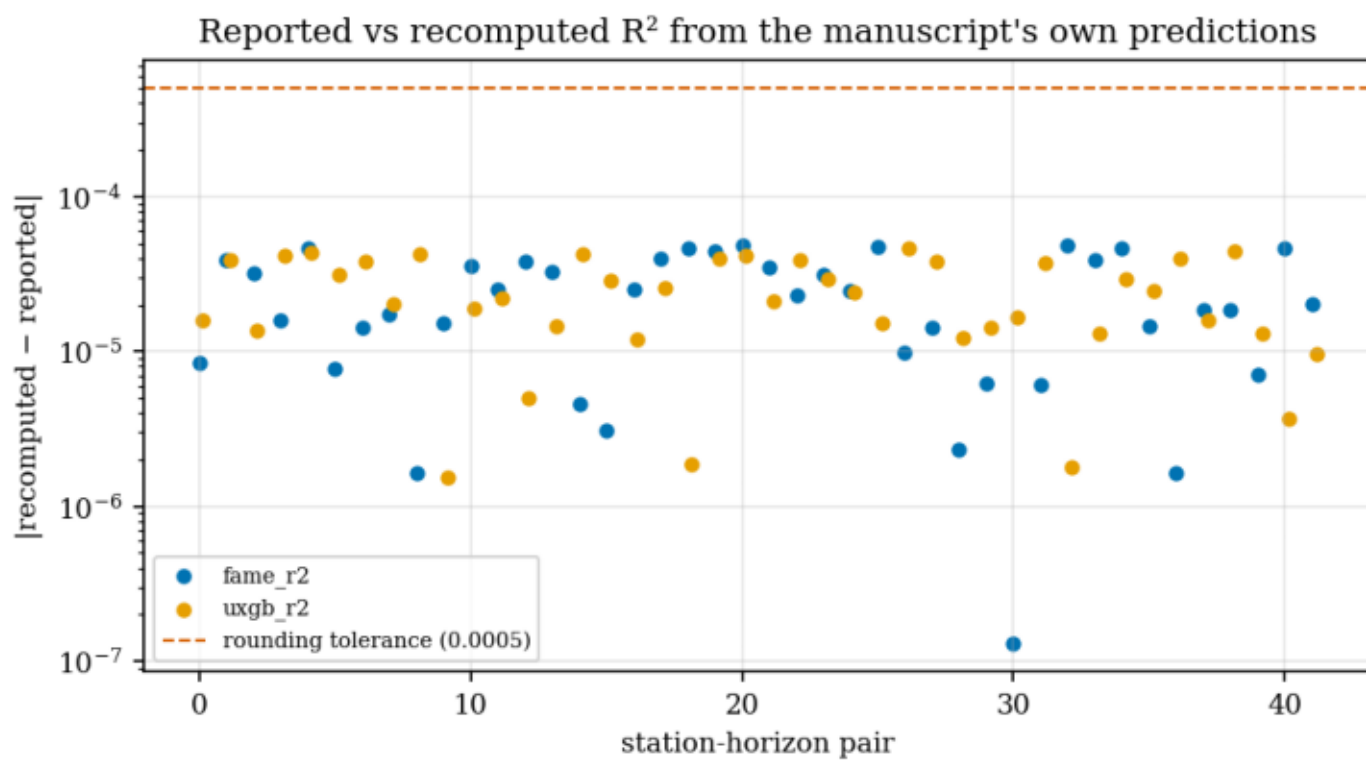
	analysis	computable_from_frozen_outputs	
Sensitivity: wavelet basis (haar/db2/db4/db6/db8/sym4/coif2)		False	changing the b
Sensitivity: trailing-window length W (128-2048)		False	
Sensitivity: decomposition level J (1-5)		False	
Sensitivity: meta-learner ridge alpha		False	the saved files store the FUSED forecast
Ablation: remove a band expert (trend/daily/hourly/noise)		False	
Ablation: causal vs whole-series decomposition		False	
Multi-seed variance of the proposed model		False	
Post-hoc fusion of saved predictions		True	
Statistical validation (Wilcoxon, Cohen's d, CIs, Holm, bootstrap)		True	
Uncertainty: split conformal on all stations and horizons		True	
Robustness, error structure, conditional skill		True	
Result consistency verification		True	

S25_computational_facts — Measured computational facts from the frozen

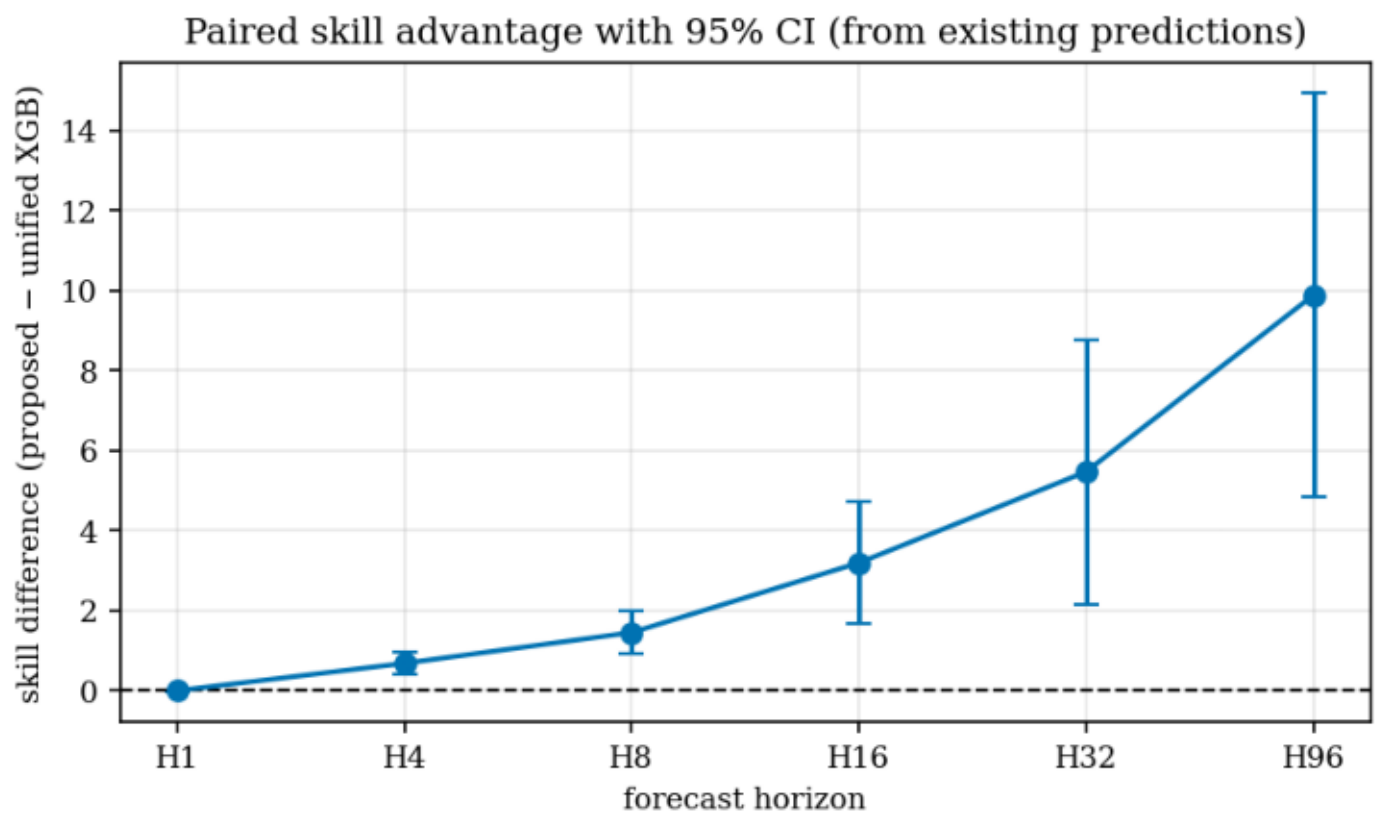
source: S25_computational_facts.csv

	quantity	value	unit	source
station-horizon pairs with saved predictions	42		pairs	results/predictions/
total stored test predictions	2.1e+05		samples	results/predictions/
models per pair	3		models	y_true, fame, unified_xgb, persistence
LSTM band expert training (exact architecture)	966		s/pair	measured; runtime_probe.json
XGBoost hourly expert training	4.944		s/pair	measured; runtime_probe.json
Ridge trend expert training	0.0093		s/pair	measured; runtime_probe.json
causal decomposition	8.085		s/station	measured; runtime_probe.json
proposed model, one pair	971		s	measured; runtime_probe.json
one 42-pair grid	11.33		h	measured; runtime_probe.json

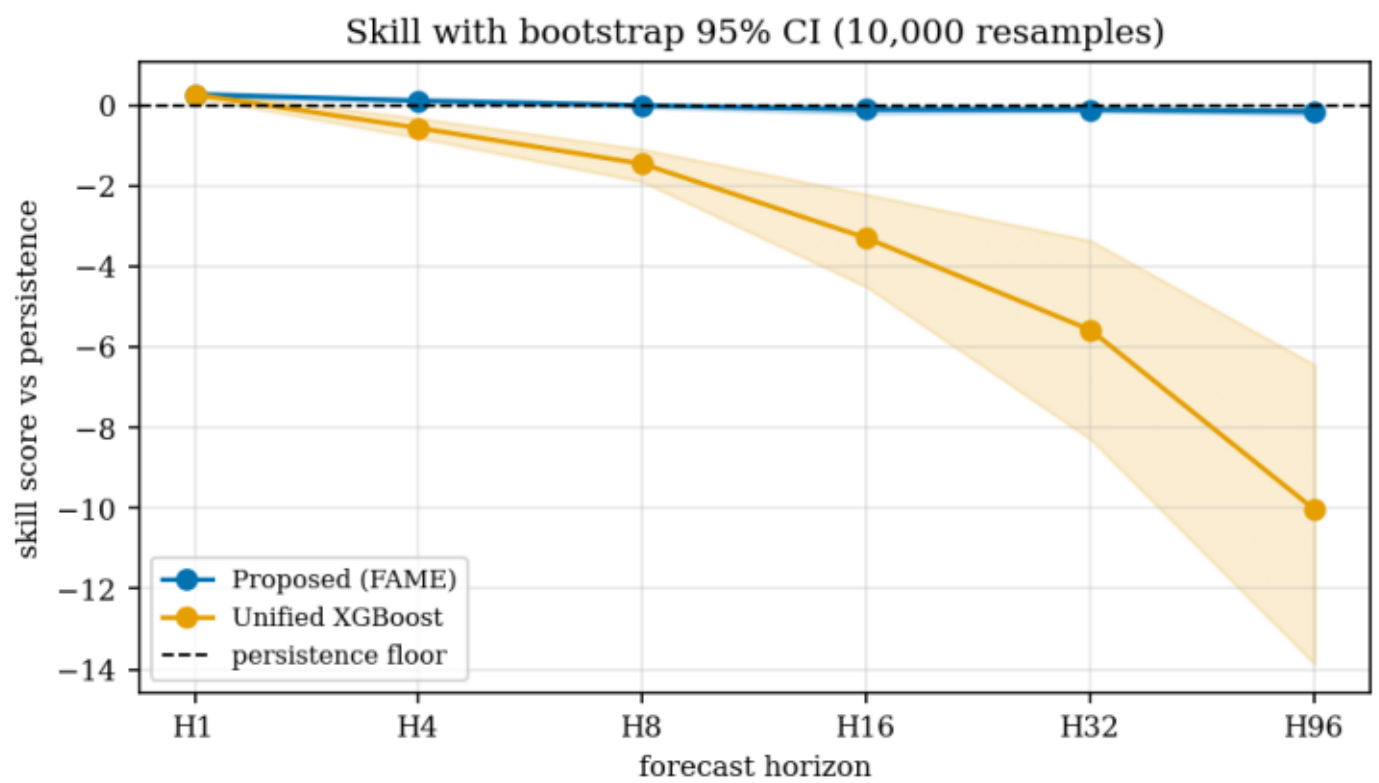
Reported vs recomputed R^2 (consistency)



Paired skill advantage with 95% CI

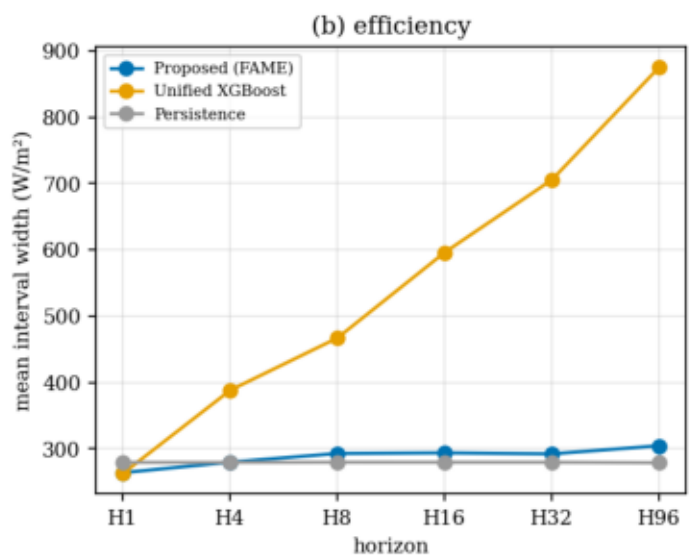
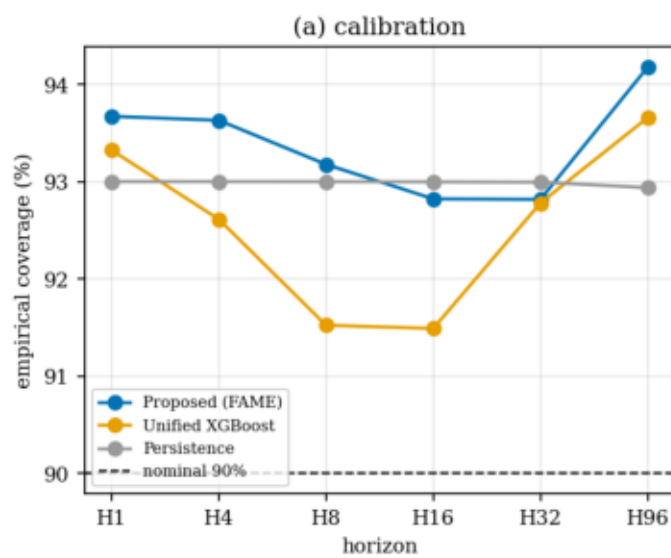


Skill with bootstrap 95% CI



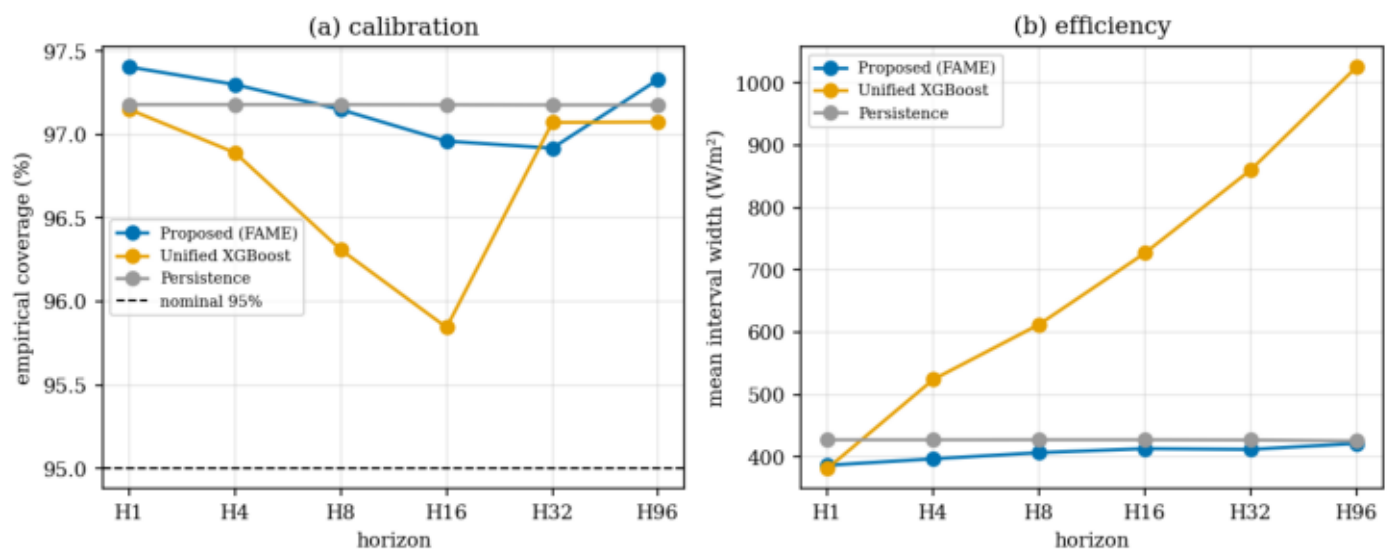
Conformal intervals at 90% nominal

Split-conformal intervals at 90% nominal — all 7 stations $\times$ 6 horizons

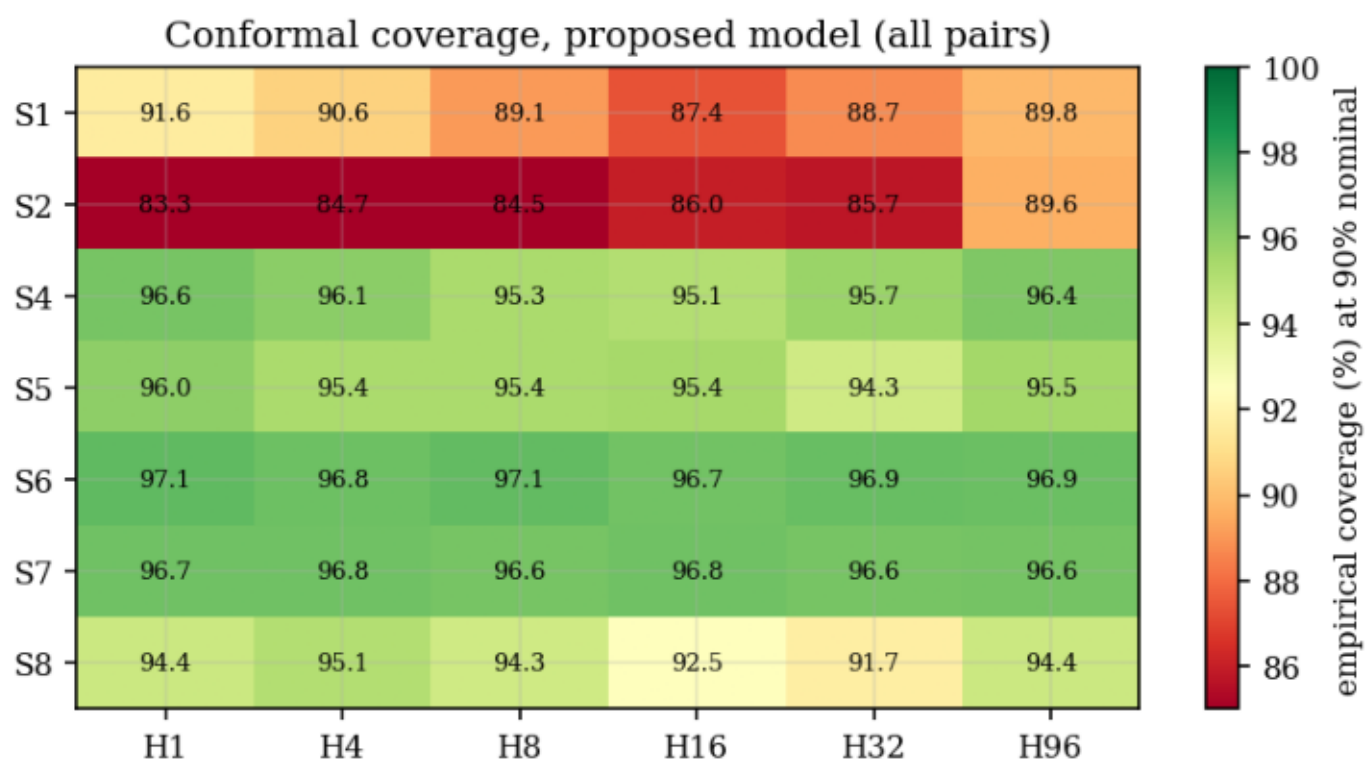


Conformal intervals at 95% nominal

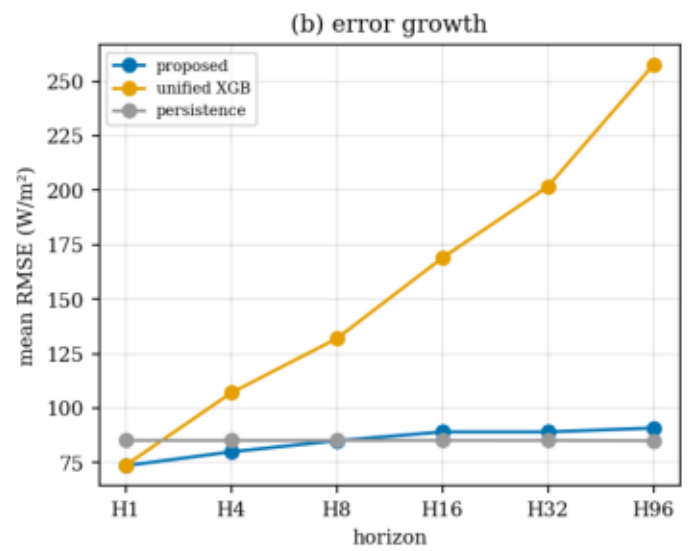
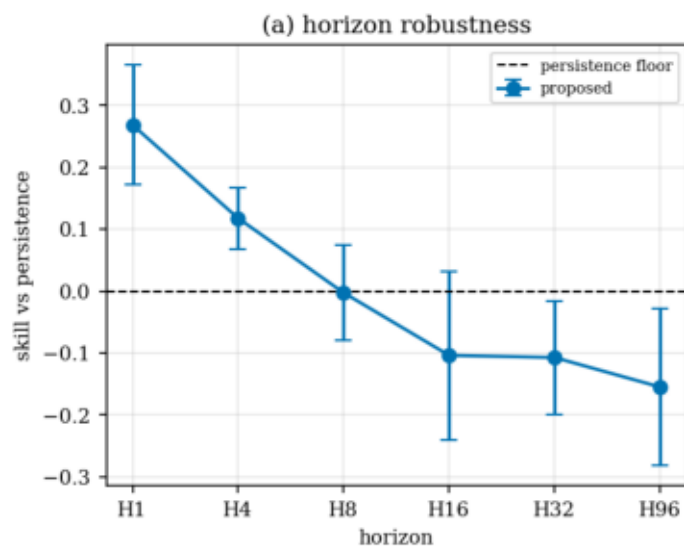
Split-conformal intervals at 95% nominal — all 7 stations $\times$ 6 horizons



Conformal coverage, all pairs

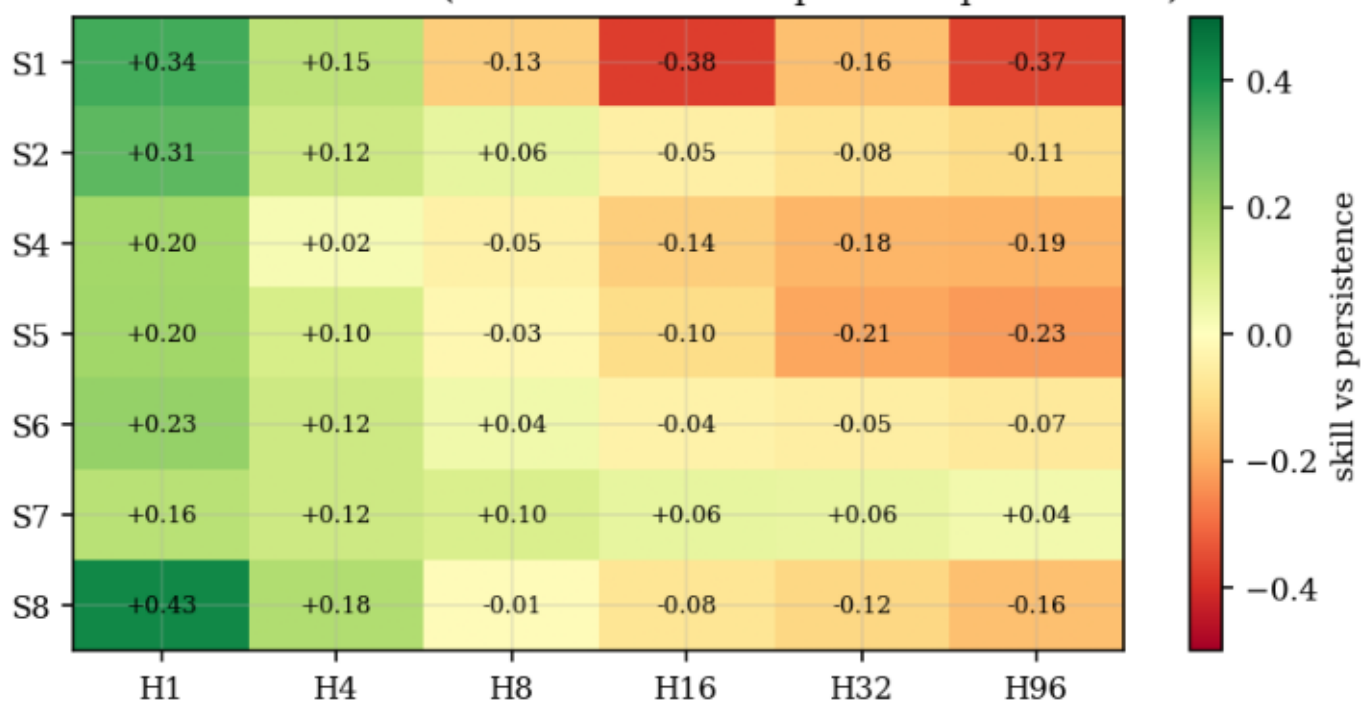


Horizon robustness and error growth



Station × horizon skill

Station × horizon skill (from the manuscript's own predictions)



Error by irradiance regime

